

Local Evidence Is Not Completion: Evidence-Condition Geometry for Bounded Completion-Audit Workflows

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Abstract

Bounded completion-audit workflows with declared source-route scopes often need to decide whether a task is complete before the remaining workload is known. Common stop signals, including no-new findings, agent agreement, stable summaries, and self-reported completion, can be useful local evidence, but they do not specify the conditions under which that evidence was produced. The key question is not whether a workflow has stopped, but whether the evidence condition under which it stopped can support the declared scope of the completion claim. We study this certificate mismatch and introduce source-route evidence-condition geometry as a runtime diagnostic. A source-route stratum records both where the workflow searched and which audit route it applied. We then define an evidence-condition controller that rejects SAFE when the current condition is too localized, repairs weak plausible strata, and returns SAFE, CONTINUE, or ABSTAIN. Across four bounded completion-audit tests, homogeneous route reuse produces localized evidence conditions and would lead to false certification if accepted as a scope-wide stop. Broader source-route exposure improves completion eligibility, while an external `urllib3` audit shows the key boundary: broad exposure is not sufficient for SAFE when residual evidence remains. The paper provides a diagnostic and control principle for bounded audit completion, not a universal completion guarantee for open-world agent workflows.

1 Introduction

This paper evaluates bounded completion-audit workflows with declared source-route scopes. These are workflows whose intended evidence region is bounded, such as a repository snapshot or document set, and whose completion claim is stated over a declared set of sources and audit routes. Examples include searching a repository snapshot for all instances of a behavior, auditing a document set for policy exceptions, checking whether an implementation covers declared failure-mode routes, or deciding whether a multi-agent audit has exhausted useful evidence within its declared scope. Such workflows must eventually stop. The difficulty is that the absence of newly found evidence is not the same object as evidence that the intended bounded workload is complete.

The failure studied in this paper is *certificate mismatch*. A workflow may produce a plausible local stop signal: repeated scans return no new findings, agents agree with each other,

subtasks appear closed, or a summary becomes stable. These signals are conditioned on the parts of the task that were searched and on the routes by which they were searched. If the workflow repeatedly applies the same route to the same subset of sources, a no-new signal may certify only local exhaustion. It does not automatically support a global claim that the task is complete over its intended scope.

Consider a code-audit workflow. Auditing a file for timeout behavior does not certify that the same file has been audited for TLS behavior, exception behavior, retry behavior, or cleanup behavior. The source may be familiar, the agents may agree, and the workflow may have stopped finding timeout-related evidence; none of that establishes that other routes over the same source have been exhausted. The stop evidence is local in its evidence condition, while the completion claim is global in scope.

We formalize this observation with *evidence-condition geometry*. The basic unit is a source-route stratum, consisting of a bounded evidence region and an audit or search route. At runtime, the workflow accumulates exposure counts over these strata. The normalized exposure distribution describes the condition under which completion evidence was produced. When this distribution is localized, stop evidence is locally conditioned. It may be useful, but it is not a global completion certificate.

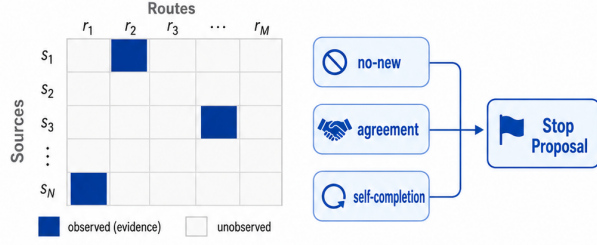
Based on this geometry, we define an evidence-condition controller. At a proposed stop, the controller computes the source-route exposure condition, checks whether the condition is broad enough to make the completion claim eligible for certification, and triggers repair over weak plausible strata when necessary. The controller is intentionally conservative: broad exposure gives completion eligibility, not automatic safety. A SAFE decision additionally requires that repair or audit reveal no residual evidence. If residual evidence appears, the controller outputs CONTINUE; if budget is exhausted without a defensible certificate, it outputs ABSTAIN.

Our experiments evaluate this controller on four bounded completion-audit task families: two generated scored tasks and two external real-repository audits with pattern-defined oracles. These tasks are not presented as a benchmark for the best item discovery policy. Instead, they test whether source-route exposure localization diagnoses unsafe stop claims and whether a controller can avoid accepting locally conditioned evidence as global completion proof.

This paper makes three contributions. First, it formulates false completion as a certificate mismatch between the declared scope of a stop claim and the source-route condition

Local Evidence Is Not a Global Completion Certificate

1 Observed Evidence Condition



2 unsafe promotion

Promoting local stop evidence to a global claim is invalid.

3 Global Completion Claim

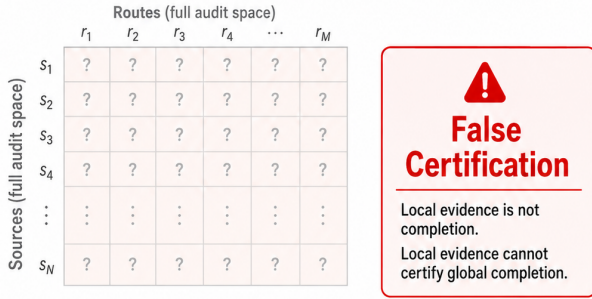


Figure 1: Certificate mismatch from local stop evidence. A workflow may observe no-new findings or agreement under localized source-route exposure. Such evidence can support local exhaustion, but it does not by itself certify completion over the intended source-route space. The evidence-condition controller audits this condition before accepting SAFE.

under which supporting evidence was produced. Second, it introduces source-route exposure geometry and an evidence-condition controller that separates completion eligibility from completion proof, repairing weak plausible strata before accepting SAFE. Third, it provides bounded audit evaluation, including generated audits and external repository audits, showing that homogeneous route reuse creates false certification risk while broader source-route evidence improves eligibility without being sufficient for safety.

We deliberately do not claim that residual-potential is an optimal repair method. It is one mechanism-aligned repair instance used to test whether weak source-route regions can expose residual evidence. The main claim is about the conditions under which completion evidence can support a stop decision.

2 Related Work

High-recall review, active search, and missing mass. Technology-assisted review and total-recall systems study how to retrieve nearly all relevant items under review bud-

gets [Grossman and Cormack, 2011, Cormack and Grossman, 2014, 2015, 2016, Grossman et al., 2016, Roegiest et al., 2015], while active search and active learning choose actions or examples to expose positives efficiently [Lewis and Gale, 1994, Settles, 2009, Garnett et al., 2012, Attenberg et al., 2015]. Missing-mass and unseen-species estimators ask how much probability mass remains unseen [Good, 1953, Chao, 1984, Chao et al., 2014]. Our scored subclasses use recall only for post-hoc false-certification measurement. The controller does not observe oracle totals, recall, or hidden missing mass; it observes the source-route condition under which a stop claim was produced and whether repair reveals residual evidence.

Agent workflows and verification. Tool-using, browser-assisted, multi-agent, and memory-based language agents allocate work across tools, agents, searches, and intermediate states [Yao et al., 2023, Schick et al., 2023, Nakano et al., 2021, Qin et al., 2024, Wu et al., 2023, Park et al., 2023, Shinn et al., 2024, Madaan et al., 2023, Wang et al., 2024]. Agent benchmarks and factual verification benchmarks evaluate task success, tool use, factuality, or evidence support [Yao et al., 2022, Deng et al., 2024, Zhou et al., 2024, Liu et al., 2024, Jimenez et al., 2024, Hendrycks et al., 2021a, Liang et al., 2022, Srivastava et al., 2023, Lin et al., 2022, Thorne et al., 2018, Gao et al., 2023, Min et al., 2023, Schuster et al., 2021]. Our question is narrower and precedes final answer scoring: whether the evidence condition under which a workflow stopped is broad enough for the scope of its completion claim.

Agreement, audit agents, and oversight. Ensembles, self-consistency, verifier-guided systems, and audit agents can improve reliability [Dietterich, 2000, Cobbe et al., 2021, Wang et al., 2023]. Agreement is not useless, but it is unsafe as an unconditioned completion certificate: agents may agree because they repeatedly examined the same local region. Safety and oversight work studies broader failure modes and scalable supervision [Amodei et al., 2016, Christiano et al., 2017, Irving et al., 2018, Leike et al., 2018, Hendrycks et al., 2021b]. This paper targets a specific operational failure within that broader landscape: a workflow may present a global completion certificate whose supporting evidence was produced under a local source-route condition.

3 Problem Setting

We study bounded completion-audit workflows with unknown residual workload and declared source-route scope. A workflow receives a task, allocates work across agents or tools, gathers evidence under a finite budget, and must decide whether the task is complete within that declared scope. Let

$$W = (T, S, R, A, B),$$

where T is the task, S is a set of sources, R is a set of routes, A is the set of agents or tools, and B is a finite budget. A

source is a bounded evidence region, such as a file, module, document, source family, or repository component. A route is an audit or search lens applied to a source.

At runtime, the workflow produces evidence events

$$e_t = (a_t, x_t, r_t, u_t, o_t, c_t),$$

where a_t is the agent or tool, $x_t \in S$ is the source, $r_t \in R$ is the route, u_t is the action, o_t is the observation, and c_t is the cost. Every evidence event is produced under a source-route condition. A source-route stratum is $s = (x, r)$, and the intended evidence-condition space is

$$\Omega = S \times R.$$

The source-route space does not require the workflow to know all true missing items in advance. It requires the intended audit scope to be defined: which sources and routes are relevant to the completion claim. This assumption is weaker than knowing the answer set and stronger than an unbounded open-world search. It matches many practical audits: a system may know which repositories, documents, modules, or evidence sources are in scope, and it may know which audit routes are relevant, while still not knowing how many true findings remain. The controller therefore evaluates the condition under which evidence was gathered, not the unknown total workload itself.

A stop claim is a workflow assertion

C_t : T is complete over the intended workload scope.

A false certification occurs when the workflow accepts SAFE for a stop claim that is not supported by the evaluation oracle:

$$\text{SAFE}(C_t) = \text{true}, \quad \text{complete}(T) = \text{false}.$$

In item-discovery subclasses, this is operationalized as accepting SAFE while bounded-oracle recall is below the completion threshold. In broader completion audits, it means accepting a global completion claim while relevant support, contradiction, or unresolved evidence remains outside the evidence condition. The oracle is used only for evaluation. It is not available to route assignment, repair scoring, eligibility checks, or stop decisions.

4 Evidence-Condition Geometry

Let $v_t(s)$ be the runtime-visible exposure count for source-route stratum s up to time t . Exposure may count visits, scans, route-specific audits, tool calls, or other logged search actions. Define

$$p_{\text{exp}}(t, s) = \frac{v_t(s)}{\sum_{s' \in \Omega} v_t(s')}.$$

This distribution is a point on the source-route coverage simplex. It describes where and how completion evidence was produced.

We use two runtime summaries:

$$\text{support}(t) = \frac{|\{s : v_t(s) > 0\}|}{|\Omega|},$$

$$G_{\text{exp}}(t) = \text{Gini}(\{v_t(s) : s \in \Omega\}).$$

These summaries are operational diagnostics, not universal laws. They summarize whether the evidence condition is localized or broad enough to make a completion certificate eligible for consideration.

Proposition 1 (Local no-new evidence is not a global certificate). *Let $U \subset \Omega$ be a strict subset of the intended source-route space. Suppose all evidence used to support a stop claim C_t is produced inside U , and suppose the workflow observes no-new evidence only for actions inside U . Then the no-new evidence can support local exhaustion over U , but it cannot by itself support global completion over Ω .*

Proof sketch. No-new evidence is conditioned on the source-route actions that were executed. If no events are produced in $\Omega \setminus U$, the workflow has not observed the absence of residual evidence in those unexposed strata. A world in which U is exhausted and $\Omega \setminus U$ contains residual evidence is observationally indistinguishable from a globally complete world under the workflow’s local evidence log. Therefore local no-new evidence rules out additional findings only under the searched condition U ; without additional assumptions about unsearched strata, it cannot rule out residual evidence in $\Omega \setminus U$.

Source-only geometry is too coarse: auditing a file for time-out behavior does not certify that TLS, retry, exception, or cleanup behavior has been audited in that file. Source-route-action geometry is more detailed, but it is sensitive to logging conventions and current experiments do not show a stable advantage over source-route. Source-route is the default because it captures both where and how evidence was produced while remaining interpretable and runtime-computable. The geometry is thus a certificate diagnostic rather than a semantic theory of all possible evidence. It asks whether the current log covers the intended source-route space broadly enough for the proposed stop claim; it does not assert that all routes are equally important or that support/Gini thresholds transfer unchanged across domains.

5 Method

The method has four modules: source-route exposure estimation, eligibility gating, residual-evidence repair, and the final SAFE/CONTINUE/ABSTAIN decision rule. The point is not to prove universal completion; it is to make the runtime evidence condition explicit before a stop claim is accepted.

5.1 Source-Route Exposure Estimation

Input: a runtime evidence log E_t , intended source set S , intended route set R , and their product space $\Omega = S \times R$. Output: exposure counts $v_t(s)$, normalized exposure $p_{\text{exp}}(t, s)$, source-route support $\text{support}(t)$, and exposure concentration $G_{\text{exp}}(t)$. Role: summarize where and how evidence was gathered, so localized stop evidence can be distinguished from broad evidence. Algorithm 1 shows how the four modules are composed at a proposed stop.

Evidence-Condition Geometry and Safe-Stopping Controller

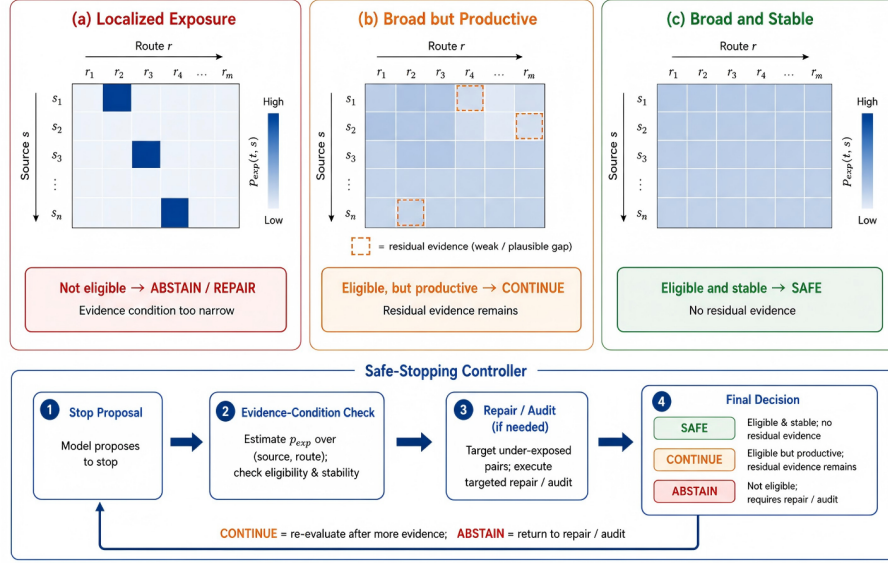


Figure 2: Evidence-condition controller. The controller maps a runtime evidence log to an exposure distribution over the intended source-route space. The support/Gini check is an eligibility test, not a completion proof. If the condition is localized or weak plausible gaps remain, the controller repairs or continues. It returns SAFE only when the evidence condition is broad and repair or audit reveals no residual evidence.

Algorithm 1 Evidence-Condition Controller

Require: $\log E_t$, space Ω , claim C_t , budget B , thresholds

τ_s, τ_g , repair rule $\mathcal{R}$

Ensure: decision in {SAFE, CONTINUE, ABSTAIN}

- 1: Compute $v_t(s)$, $p_{\text{exp}}(t, s)$, $\text{support}(t)$, and $G_{\text{exp}}(t)$
- 2: $\text{eligible} \leftarrow [\text{support}(t) \geq \tau_s \wedge G_{\text{exp}}(t) \leq \tau_g]$
- 3: **if** not *eligible* and $B = 0$ **then**
- 4: **return** ABSTAIN
- 5: **end if**
- 6: **if** not *eligible* or weak plausible gaps remain **then**
- 7: Audit $Q = \mathcal{R}(E_t, \Omega, C_t)$ within B
- 8: **if** residual evidence appears **then**
- 9: **return** CONTINUE
- 10: **end if**
- 11: Recompute *eligible*
- 12: **end if**
- 13: **if** *eligible* and no residual evidence appeared **then**
- 14: **return** SAFE
- 15: **end if**
- 16: **return** ABSTAIN

The source-route exposure summaries are operational diagnostics, not universal laws. They summarize whether the evidence condition is localized or broad enough to make a completion certificate eligible for consideration.

5.2 Eligibility Gate

Input: $v_t(s)$, $p_{\text{exp}}(t, s)$, $\text{support}(t)$, $G_{\text{exp}}(t)$, thresholds τ_s and τ_g . Output: eligible / ineligible. Role: decide whether the

current evidence condition is broad enough for a stop claim to be considered.

The eligibility gate is not a completion proof. It is only a test of whether the evidence condition is broad enough for a global stop claim to be considered. In the current implementation, support and Gini thresholds instantiate this operational gate; they are not claimed as universal laws.

5.3 Residual-Evidence Repair

Input: current evidence log E_t , evidence condition Ω , claim C_t , budget B , and repair rule $\mathcal{R}$. Output: repaired log or continued audit targets. Role: probe weak plausible strata for residual evidence when eligibility alone is insufficient. Weak plausible gaps are strata that are both under-exposed in the current source-route condition and still plausible under non-oracle runtime signals such as source text, route names, route match counts, source size, or ledger-visible agent notes.

The repair instance used in the experiments is

$$\text{priority}(s) = \text{under_exposure}(s) \cdot \text{runtime_potential}(s).$$

The under-exposure term gives priority to weak parts of the current completion certificate. The runtime-potential term uses visible information such as source text, route names, source size, route match counts, or non-oracle ledger signals. It is not allowed to use oracle labels, oracle totals, oracle missing mass, undiscovered true item counts, post-hoc recall, or scorer-visible target distributions. Residual-potential is a mechanism-aligned repair instance, not an optimal active-search algorithm.

5.4 SAFE/CONTINUE/ABSTAIN Decision Rule

Input: eligibility result and residual-evidence repair outcome. Output: SAFE, CONTINUE, or ABSTAIN. Role: accept a stop only when the evidence condition is broad and repair or audit reveals no residual evidence.

The final rule is intentionally conservative. Broad exposure gives completion eligibility, not automatic safety. A SAFE decision additionally requires that repair or audit reveal no residual evidence. If residual evidence appears, the controller outputs CONTINUE; if budget is exhausted without a defensible certificate, it outputs ABSTAIN.

6 Experiments

The experiments are not a discovery benchmark. They are controlled completion-audit tests designed to check whether stop evidence is sufficient to support a completion claim under a declared source-route scope.

We organize experiments around the method design points. First, do naive stop signals create false certification? Second, is source-route geometry necessary relative to source-only exposure? Third, is eligibility alone sufficient? Fourth, does the full controller avoid unsafe stops without becoming a never-stop rule? Fifth, how does residual-potential repair compare with random and high-potential repair?

Tasks. `policy_docset_v1` is a generated policy document set with four sources covering access control, data handling, release process, and audit exceptions. Its routes are obligation, exception, deadline, and prohibition; the hidden oracle contains 24 policy clauses. `code_repo_v1` is a generated bounded code repository with authentication, payment, storage, and API-client sources. Its routes are compatibility, security, and resilience; the hidden oracle contains 20 code-risk items.

`requests` is an external real-repository audit over six files from a local installed snapshot of the Python package: adapters, API, authentication, models, sessions, and utilities. Its routes are TLS, timeout, exception, and compatibility. `urllib3` is an external real-repository audit over connection, connection-pool, pool-manager, response, retry, and timeout sources. Its routes are timeout, retry, TLS, exception, and cleanup, giving 30 intended source-route strata. External oracles are pattern-defined from frozen snapshots: stronger than generated toys, weaker than human-annotated completion-audit benchmarks.

Workflow conditions. Homogeneous route reuse assigns multiple agents to the same route across sources. The reused routes are obligation for `policy_docset_v1`, compatibility for `code_repo_v1`, and timeout for the two external repository audits. Route-partitioned audit assigns agents to different routes over the same source set. `urllib3` additionally includes an `extended_audit` condition covering all five routes.

Compared policies and ablations. We compare decision variants and repair-target variants on fixed bounded-audit states. Decision variants are Naive stop, Source-only, Source-route eligibility-only, and the full controller. Repair-target variants keep the full decision rule fixed and choose strata by Random, High-potential, or Residual-potential. All policies use the same fixed trajectories and proposed stop states; differences come from the decision rule or repair target selection, not from different discovery trajectories. Oracle labels are used only for post-hoc scoring.

Seeds, budgets, thresholds, and cost. Generated tasks use budget 4; `requests` and `urllib3` use 200 seeds with budgets 4 and 5, respectively. The controller uses $\tau_s = 0.75$, $\tau_g = 0.70$. A 0.90 recall threshold is used only for bounded-oracle false-certification evaluation and is invisible to the controller. Repair cost counts only additional scanned source lines plus extraction events after a stop state is fixed. Seeds, route assignments, repair orders, and challengers are fixed before oracle scoring.

The four task families supply bounded completion-audit cases with withheld or post-hoc oracles. Simulation sweeps are screening checks, not new task claims. Leakage controls are in the supplement; runtime decisions use visible logs and source text, not oracle labels or undiscovered true-item counts.

7 Results

Naive stop signals cause false certification. Figure 3 begins with the failure mode. Across all four tasks, homogeneous route reuse produces localized evidence and would falsely certify completion if accepted as SAFE. Controlled sweeps show the same pattern: homogeneous and prompt-diverse conditions have near-complete false-stop failure. This motivates source-route granularity rather than accepting unconditioned stop signals.

Source-route geometry is necessary. The middle panel of Figure 3 is the source-only vs. source-route ablation. Homogeneous runs touch all files, so source-only support looks complete; the same states have localized source-route support and sub-threshold recall. Source-only exposure cannot detect that timeout evidence does not certify TLS, retry, exception, or cleanup routes in the same file. This is the source-route granularity design point.

Eligibility alone is not proof. Route-partitioned evidence improves eligibility, but it is not proof. In `urllib3`, route-partitioned exposure covers 24/30 strata and passes the gate, yet recall remains below threshold and repair finds residual evidence; the controller returns CONTINUE, not SAFE. The extended audit reaches support and recall 1.000. This is the eligibility/proof separation design point.

Full controller avoids unsafe stops without never-stopping. A naive controller falsely certifies homogeneous stops; the

Table 1: Experimental coverage. Oracles are used only for post-hoc scoring.

Study	Sources	Routes	Oracle items	Runs/seeds	Main measurement
policy	4	4/16	24	200/challenger	FCR, repair gain
code	4	3/12	20	200/challenger	FCR, repair gain
requests	6	4/24	298	200/challenger	controller validation
urllib3	6	5/30	699	200/challenger	boundary validation
Controlled geometry	–	variable	synthetic	2160	metric screening
Exposure localization	–	18/30/42	synthetic	2700	localization check

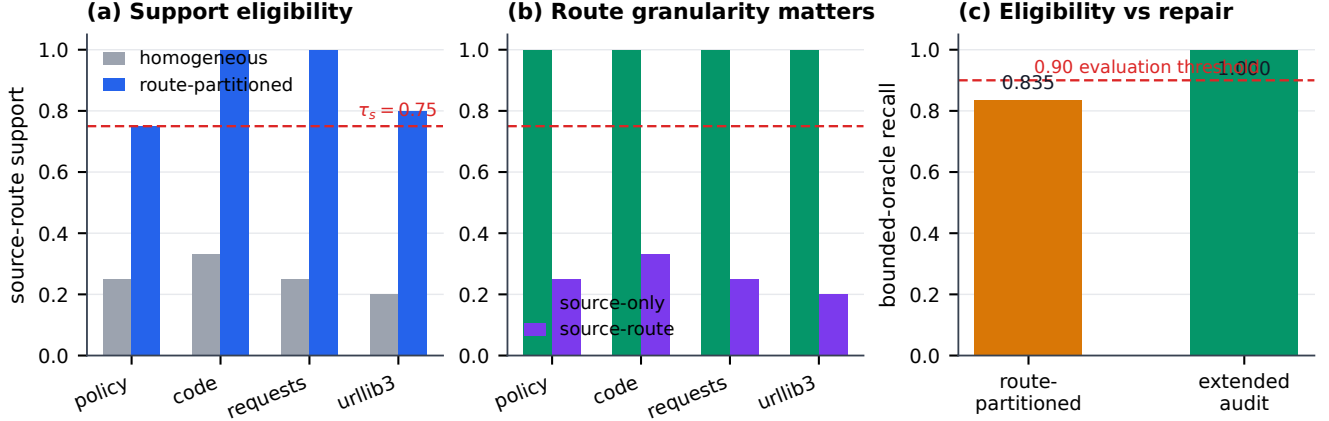


Figure 3: Main diagnostic and ablation. Left: broader audits increase source-route support. Middle: source-only support is full in homogeneous runs, but source-route support remains localized. Right: `urllib3` as eligibility-only vs. eligibility+repair; broad route-partitioned exposure is still incomplete, while the extended audit reaches completion.

full controller rejects them. It is not a never-stop rule: it returns SAFE for broad complete states in the generated tasks, `requests`, and the extended `urllib3` audit. In seeded external validations, repair states are oracle-unsafe and return CONTINUE or ABSTAIN, while 6000 oracle-safe complete state perturbations keep safe coverage 1. This is the non-never-stop decision-behavior design point.

Repair policy comparison. Figure 4 separates repair target variants from decision variants. Residual-potential exposes residual evidence and has higher mean gain than random repair, but it is not proven optimal; high-potential is competitive in cost-normalized terms. Repair is therefore a bounded probe for remaining evidence, not a claim of optimal active search.

Sensitivity and audit checks. Sensitivity sweeps are in the supplementary material. Varying τ_s, τ_g and repair budget changes the SAFE/CONTINUE/ABSTAIN mixture and repair cost, but does not create false certifications in the external controller validations. Oracle construction, negative controls, leakage checks, seeded details, and regex examples are also supplementary.

8 Discussion and Validity

The controller certifies a relationship between a stop claim and the evidence condition that produced its support. It does not certify that the world contains no further facts, that future searches would fail, or that source-route is the only possible abstraction. Recall is used only for post-hoc measurement; runtime decisions use exposure, eligibility, and repair outcomes. The main threat is oracle leakage, controlled by freezing trajectories before scoring and limiting runtime features to visible logs, source text, lexical counts, route identifiers, and non-oracle ledger signals. Thresholds are operational test points, not universal constants.

9 Limitations and Supplement

The evidence supports a diagnostic and controller for bounded completion-audit subclasses. External oracles are pattern-defined; source-route strata are task-designed; and residual-potential is not proven optimal. The supplement contains oracle construction, negative controls, leakage checks, seeded safe-state details, route examples, and reproducibility material.

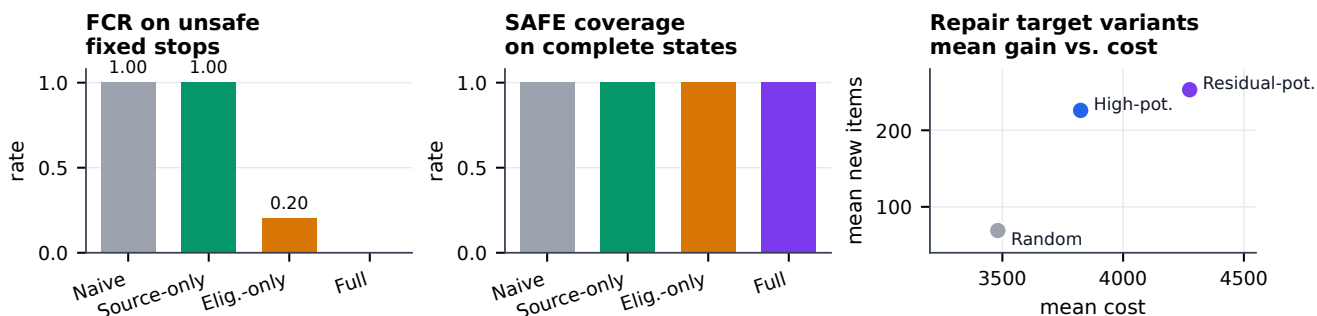


Figure 4: Controller variant comparison. Left/middle: decision variants on the same fixed stop-state set. FCR is the micro-average over oracle-unsafe fixed stops; safe coverage is the micro-average over oracle-safe fixed stops and seeded complete states. Right: repair target variants keep the full decision rule fixed and compare mean repair gain vs. mean post-stop repair cost over oracle-unsafe repair states.

10 Conclusion

Local evidence does not automatically support a scope-wide completion claim. Source-route evidence-condition geometry makes this conditioning explicit. The controller uses it to reject unsafe local stops, repair weak plausible strata, and reserve SAFE for broad evidence conditions with no residual evidence. The contribution is a bounded diagnostic and control principle: completion certificates should be judged by the evidence condition that produced them.

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